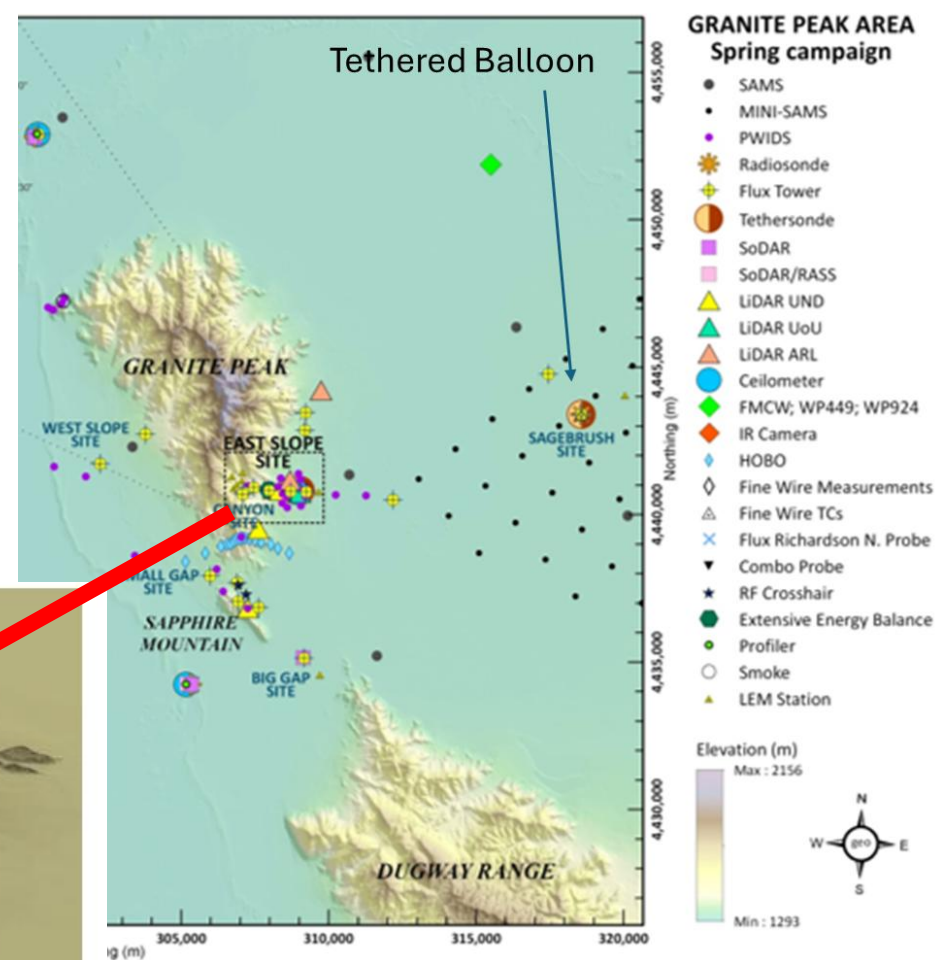
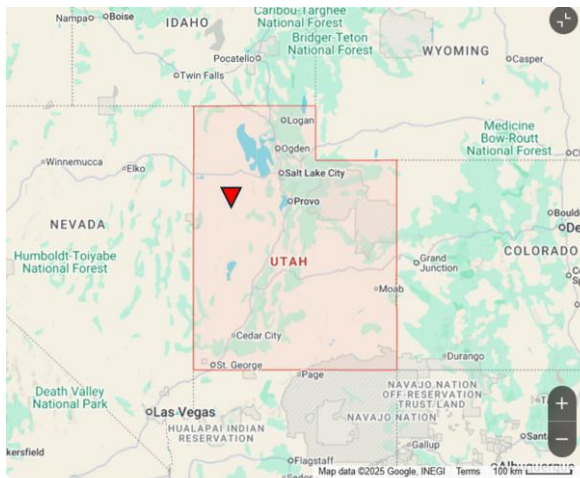
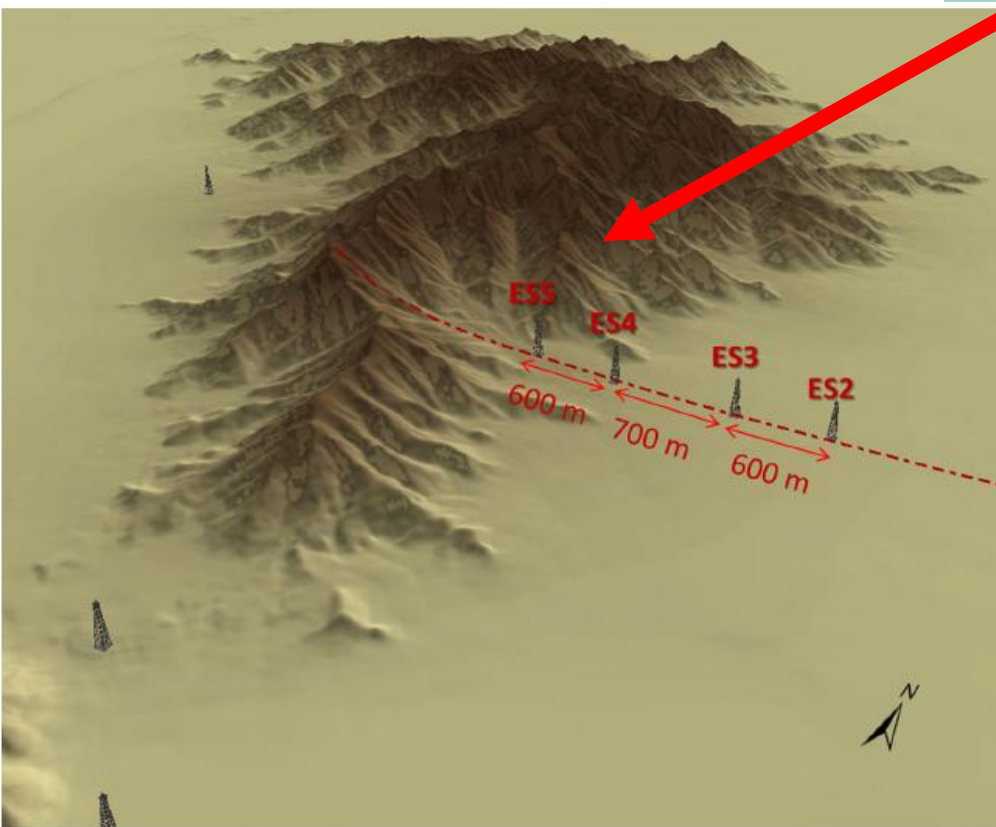


Hands-on: Analysis of Sonic Data collected at a slope site



Granite Peak: isolated topography with length 11.8 km, the largest width is 6.1 km, and peak elevation 0.84 km above the valley elevation (1.3 km above sea level).



DATASET

ES5 Tower Instrumentation:

Sonic Anemometer [0.5; 2; 5; 10; 20] m

Thermo-hygrometer [0.5; 2; 5; 10; 20] m

Barometer [0.5] m

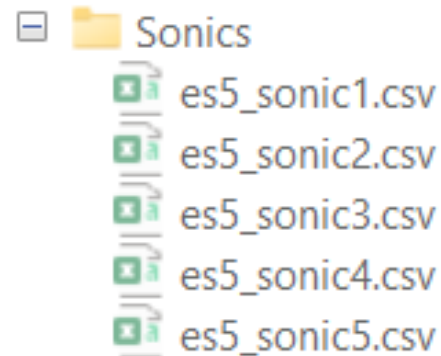
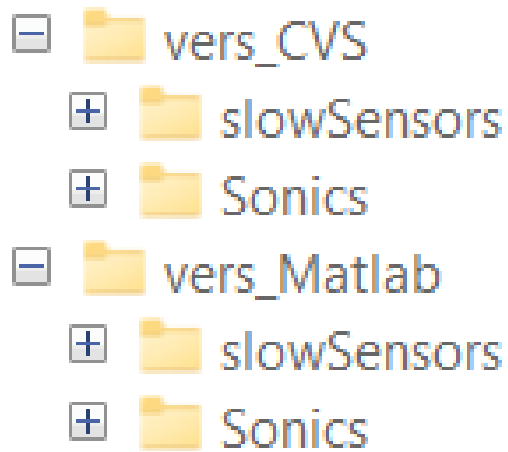
ES5 Tower Location

	Lon	Lat	Elevation (m AGL) - Slope angle [°]	
ES5	-113,25861	40,09652	1433	3,6

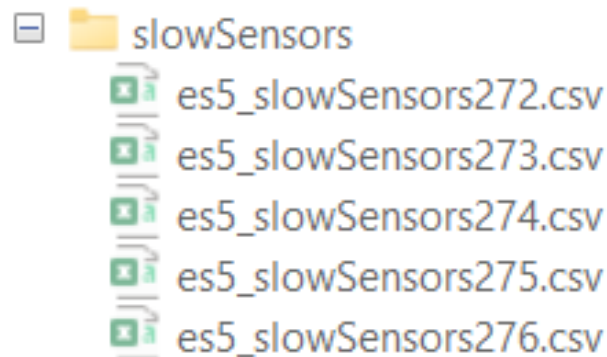
Dataset to be analyzed (clear sky conditions, weak synoptic forcing):

- Multiple days starting from Sept 28, 2012
- Sonic Data and other sensors already averaged (note the different average time between sonic vs slow sensors)
- Double-Rotation applied

DATASET



One file per level of measurement:
sn1= lowest level (0.5 m)
....
sn5= top level (20m)



One file per day (272-276 JD)
containing all levels
(from 1= lowest level (0.5 m)
to 5= top level (20m)

DATASET -SONIC

Time info

Speed

Dir

Ts
(sonic
Temp.)

$\overline{u'w'}$

$\overline{v'w'}$

$\overline{u'Ts'}$

$\overline{w'Ts'}$

$\overline{u'Ts'}$

	A	B	C	D	E	F	G	H	I	J	K
1	jday_sn1	hour_sn1	minx_sn1	wsp_sn1	truewdir_s	Tsm_sn1	wu_cov_sn	wv_cov_sn	uv_cov_sn	wT_cov_sn	uT_cov_sn
2	272	0	0	0.89681	34.08266	23.88057	-0.06612	-0.00243	0.025715	0.003197	-0.00699
3	272	0	5	0.877724	34.63422	23.66745	-0.05043	0.000463	-0.00634	-0.0034	0.009845
4	272	0	10	0.854605	46.18889	23.55805	-0.05954	-0.00478	0.016106	-0.00965	0.027123
5	272	0	15	0.694901	15.16448	23.42432	-0.02165	0.007074	0.010755	-0.00825	0.020168
6	272	0	20	0.612158	39.95961	23.2994	-0.02403	-0.0032	0.004926	-0.01229	0.025085

$\overline{v'Ts'}$

σ_u

σ_v

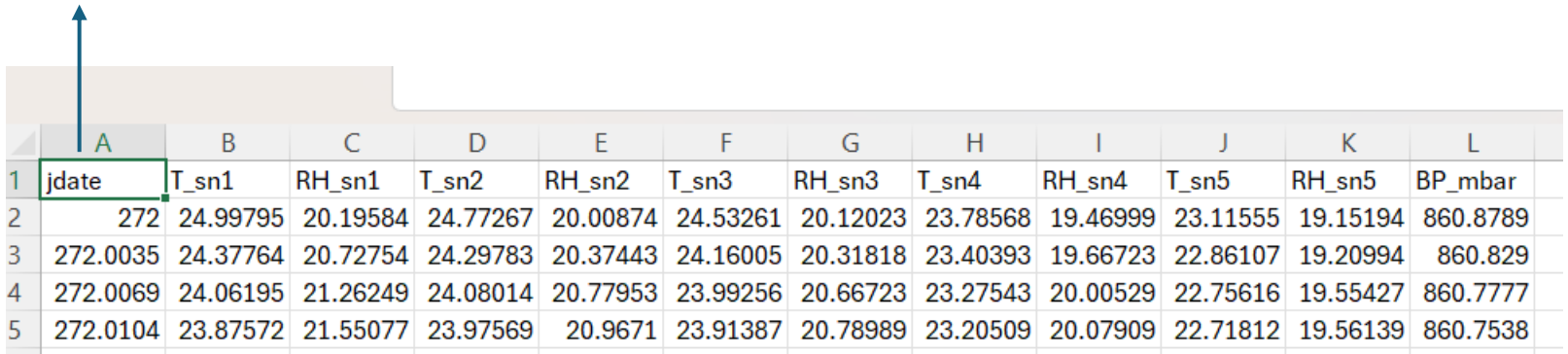
σ_w

σ_{Ts}

L	M	N	O	P	Q
wT_cov_sn1	sigu_cov_s	sigv_cov_s	sigw_cov_s	sigT_cov_sn1	
-0.0023	0.497	0.470683	0.262662	0.103573	
0.00255	0.48344	0.445296	0.231234	0.098006	
0.004853	0.499004	0.431772	0.249483	0.108911	
0.009169	0.347306	0.30834	0.185776	0.133215	
0.005996	0.329257	0.273611	0.172665	0.15847	

DATASET –Slow Sensors

Time info (note the different format compared to sonic data)



	A	B	C	D	E	F	G	H	I	J	K	L
1	jdate	T_sn1	RH_sn1	T_sn2	RH_sn2	T_sn3	RH_sn3	T_sn4	RH_sn4	T_sn5	RH_sn5	BP_mbar
2	272	24.99795	20.19584	24.77267	20.00874	24.53261	20.12023	23.78568	19.46999	23.11555	19.15194	860.8789
3	272.0035	24.37764	20.72754	24.29783	20.37443	24.16005	20.31818	23.40393	19.66723	22.86107	19.20994	860.829
4	272.0069	24.06195	21.26249	24.08014	20.77953	23.99256	20.66723	23.27543	20.00529	22.75616	19.55427	860.7777
5	272.0104	23.87572	21.55077	23.97569	20.9671	23.91387	20.78989	23.20509	20.07909	22.71812	19.56139	860.7538

T_sn1=
temperature
at level 1

RH_sn1=
relative
humidity at
level 1

pressure

Objectives:

Part A:

- Identify if & when you have a downslope flow
- For such regime, characterize the vertical profiles of mean & turbulent variables of interest.
- Evaluate whether [some](#) of the theoretical concepts seen in our lectures are verified or not (*extra points* if you are able to argument the above)

Part B:

Give an overview of the overall patterns observed during the entire observational period (recall that you have multiple days)

PART C:

Suppose you have (limited) budget to do a second campaign :
what instrument set- up will you choose in order to

- 1) better verify the conclusions you are able to draw from PartA and/or B;
OR (/and)
- 2) to further explore or validate the interpretative hypotheses you may have developed based on Part A and/or Part B.

[Some= just focus on few aspects, you do not to cover everything](#)